

RANDOM FOREST Classifier

confusion matrix Assignment:

1.Accuracy - What is the % of correctly classified purchase yes and no from the total test data?

$$TP+TN/TP+TN+FP+FN = 79+45/79+6+4+45 = 124/134 = 0.9253$$

2.Recall - Purchase_yes(what is the % of correctly classified purchase yes from the total purchased yes from test data?)

$$TP/TP+FN = 79/79+6 = 0.9294$$

Purchase_no(what is the % of correctly classified purchase No from the total purchased no from test data?)

$$TN/TN+FP = 45/45+4 = 0.9183$$

3.Precision - Purchase_yes(what is the % of correctly classified purchase yes to sum of correctly classified purchase yes and wrongly classified purchased yes from test data?)

$$TP/TP+FP = 79/79+4 = 79/83 = 0.9518$$

Purchase_No(what is the % of correctly classified purchase No to sum of correctly classified purchase No and wrongly classified purchased No from test data?)

$$TN/TN+FN = 45/45+6 = 0.8823$$

4. F1 Score - Purchase_yes(What is overall performance of Purchase_yes?)

$$2(\text{precisionYes} * \text{RecallYes}) / (\text{precisionYes} + \text{RecallYes}) = 2(0.95 * 0.93) / (0.95 + 0.93) = 1.767 / 1.88 = 0.9398$$

Purchase_No(What is overall performance of Purchase No?)

$$2(\text{precisionNo} * \text{RecallNo}) / (\text{precisionNo} + \text{RecallNo}) = 2(0.91 * 0.88) / (0.91 + 0.88) = 1.6016 / 1.79 = 0.8947$$

5. Macro Average:

Precision - What is average performance of precision correctly and wrongly classified Purchase_yes and Purchase_No?

$$\text{Precision}(\text{yes}) + \text{Precision}(\text{No}) / 2 = 0.95 + 0.88 / 2 = 0.92$$

Recall - What is average performance of Recall correctly and wrongly classified Purchase_yes and Purchase_No?

$$\text{Recall}(\text{yes}) + \text{Recall}(\text{No}) / 2 = 0.93 + 0.92 / 2 = 0.925$$

F1 Score - What is average performance of F1 Score correctly and wrongly classified Purchase_yes and Purchase_No?

$$\text{F1 Score}(\text{yes}) + \text{F1 Score}(\text{No}) / 2 = 0.9398 + 0.8947 / 2 = 1.8345 / 2 = 0.91725$$

6. Weighted Average:

Precision - What is sum of product of proportion rate of each class Purchase_yes and Purchase_No?

$$\text{Precision}(\text{yes}) * (85/134) + \text{Precision}(\text{No}) * (49/134) = 0.95 * 85/134 + 0.88 * 49/134 = 0.60 + 0.32 = 0.92$$

Recall - What is sum of product of proportion rate of each class Purchase_yes and Purchase_No?

$$\text{Recall}(\text{yes}) * (85/134) + \text{Recall}(\text{No}) * (49/134) = 0.5895 + 0.3357 = 0.925$$

F1 Score - What is sum of product of proportion rate of each class Purchase_yes and Purchase_No?

$$\text{F1 Score}(\text{yes}) * (85/134) * (85/134) + \text{F1 Score}(\text{No}) * (49/134) = 0.596 + 0.327 = 0.923$$